University of St.Gallen

Course and Examination Fact Sheet: Autumn Semester 2025

9,005: RPV: Aviation and Space Industry

ECTS credits: 4

Overview examination/s

(binding regulations see below)
decentral - Written work, Digital, Group work group grade (80%)
Examination time: Term time
decentral - Presentation, Analog, Group work group grade (20%)
Examination time: Term time

Attached courses

Timetable -- Language -- Lecturer
9,005,1.00 RPV: Aviation and Space Industry -- English -- Puls René

Course information

Prerequisites

None.

General interest in NewSpace, Aerospace, Aviation, emerging technologies and/or business models with vertical integration is an asset. Fascination with Aviation and Space industry and/or start-ups, spin-offs and alike.

Learning objectives

By the end of the course, students will:

- Understand the strategic and operational dynamics of aerospace and NewSpace industries.
- Apply business development tools to real-world technology commercialization challenges.
- Formulate actionable business models and go-to-market strategies for high-tech offerings.
- Conduct industry and stakeholder analyses, including customer segmentation and pricing models.
- Translate technical capabilities into marketable services and solutions.
- Deliver professional-grade consulting outcomes with academic and practical rigor.

Content

Two corporate partners / projects are available (assignment will be done as part of the kick-off meeting). In 2025, RPV Aviation & Space will focus on space driven projects by one of the emerging New Space actors - **Starlab Space** ... and by the **European Space Agency**.

Project 1: Starlab Space – Commercializing Space-Based Biotech through Integrated Business Models

Corporate Partner: ORBIT FOR LIFE (Working Title)
Focus Area: Strategic Business Development for In-Orbit Biotech Innovation

Company Overview
Starlab Space (a Joint Venture by Voyager Space, Airbus, Mitsubishi Corporation, and MDA Space) is pioneering the next generation of space infrastructure (building the commercial successor of the International Space Station) by enabling **commercial research, manufacturing, and innovation in microgravity**. Together with the ORBIT FOR LIFE initiative, Starlab aims to commercialize space-based biotechnology and life sciences, leveraging microgravity environments to unlock new frontiers in pharmaceutical development, organoid engineering, and protein crystallization.

Project Objective
The student consulting project will support ORBIT FOR LIFE in **developing actionable, commercially viable business models** for in-orbit biotech ventures. Students will explore how advanced space-based research and manufacturing capabilities can be translated into sustainable commercial offerings, with a focus on maximizing return on investment (ROI), optimizing resource utilization, and aligning scientific innovation with early market adoption.

Strategic Focus Areas (selection will be done during kick-off and project plan phase)

1.Integrated Business Model Development

- Design and validate end-to-end business models combining in-orbit protein research, organoid development, and microgravity-based manufacturing.
- Develop strategies for optimizing cargo return capacity and platform utilization to enhance profitability and operational efficiency.

2.Market Research and Customer Discovery

- Conduct targeted market research on early adopters within pharma and biotech sectors.
- Perform stakeholder interviews to apply real-world customer discovery methods.
- Define ideal customer profiles, key pain points, and product–market fit criteria.

3.Funding and Investment Strategy

- Map the funding ecosystem including public and private sources, venture capital, and government grants.
- Evaluate financing timelines, capital requirements, and scale-up milestones required for biotech commercialization from space-based platforms.

4.Cross-sector Synergies and Enabling Technologies

- Analyze the convergence between life sciences and digital technologies.
- Assess innovations such as algorithmic scheduling, AI-driven experiment planning, and capacity optimization tools to boost scalability and efficiency.

5.Communication and Investor Engagement

- Produce persuasive business cases and communication plans tailored for investors, policy makers, and industry decision-makers.
- Emphasize clarity, value proposition, and expected ROI to accelerate buy-in and capital attraction.

Key Deliverables

- Comprehensive Business Model Canvases for Space-Based Biotech Products.

- Stakeholder Mapping and Interview Insights Summary.

- Investment Roadmap and Go-to-Market Strategy.

- Presentation of Business Opportunity to a Non-technical Executive Audience.
- Final Consulting Report including Financial Viability and Scaling Recommendations.

Project 2: ESA Future Talents Initiative – Building the Business Model for Europe's Space Entrepreneurship Platform

Corporate Partner: European Space Agency (ESA) – Commercialisation Services
Focus Area: Business Model Development and Strategic Planning for Space Talent Enablement

Project Context
The European Space Agency is advancing a strategic initiative to shape the next generation of European space entrepreneurs and professionals through the **Commercialisation Future Talents Initiative (FTI)**. This initiative responds to critical talent pipeline challenges in the space sector, aiming to connect tertiary education, professional development, and entrepreneurial activity via a scalable digital platform.

A cornerstone of this initiative is the **eLearning platform for space entrepreneurship**, which will offer certified, modular learning pathways to students, researchers, and professionals: bridging business, academia, and space innovation.

Project Objective

The student consulting team will support ESA in **designing and refining the business model** and **commercialization strategy** for this pan-European space entrepreneurship education platform. The focus lies in aligning multi-stakeholder incentives, platform sustainability, and value creation across universities, professionals, and institutional partners.

Strategic Focus Areas

1.Platform Business Model Design

- Develop a multi-sided business model for the eLearning platform, incorporating key stakeholders: universities, learners, ESA, ECSECO, and industry contributors.
- Define the value proposition for each segment and outline mechanisms for content certification, revenue sharing, and platform governance.

2.Market and Stakeholder Analysis

- Conduct an ecosystem mapping of academic, governmental, and industry actors active in space entrepreneurship education.
- Benchmark against global best practices (e.g., executive education, digital learning in tech sectors) to identify differentiators and positioning levers.

3.Revenue & Incentive Structure

- Propose tiered pricing models for different certificate programs (e.g., Space Entrepreneur, Business Developer).
- Assess how ECSECO can act as an education broker and value steward, ensuring quality and comparability while supporting institutional autonomy.

4.Outreach & Growth Strategy

- Develop recommendations for marketing and outreach to attract early participants and institutional partners.
- Propose mechanisms to promote adoption across Europe (including accreditation transfer, university onboarding, and incentives for module creation).

5.Scalability and Sustainability Plan

- Define a roadmap for scaling the platform across countries, institutions, and new verticals (e.g., green space tech, AI in space ops).
- Explore the long-term evolution of Space commercialisation actors: from platform manager to certification authority and innovation accelerator.

Key Deliverables

- ESA-Led Business Model Canvas and Value Chain Map
- Stakeholder Engagement & Incentive Framework
- Revenue Model and Pricing Recommendations
- Communication & Positioning Strategy for Initial Rollout
- Final Consulting Report with Strategic Options and Risk Assessment

Structure and indications of the learning and teaching design

- This course accounts for 4 credits. Accordingly, the total workload for students is 120 hours. This includes self-study, campus time and final presentation.
- The structure of the contact study is planned as follows: 2 hours of introductory lectures and 4 hours (meetings) of personal coaching and 14 hours of interactive contact in a blended learning format.
- The structure of the self-study is intended as follows: 20 hours of preparation time for the meetings with the lecturer and the practice partner and 30 hours for the examination presentation preparation and final presentation and 50 hours in blended learning specifically group work on the project.
- The course is conducted partly in presence in St. Gallen if possible for the kick-off meeting in presence for the bilateral meetings between the student team and the academic coach (presence and/or online) and practice partner (presence and/or online).

Further explanation of the teaching and learning design:
This course is set up as a consulting project course, where students will receive a briefing from the academic coach concerning project management and the project semester planning including pre defined meeting dates with the academic coach. Furthermore, students receive a kick-off briefing from the corporate partner about the project. Students then work independently (in groups) on the project after having agreed on a project proposal. Students will then have a 25%, 50 % and 90% internal meeting with the academic coach and regular meetings with the corporate partner. After the internal rehearsal (presentation dry run of 100% version with the academic coach), students will present and deliver the project results and report to the client. For this presentation it is planned to visit the client.
The structure is as follows:

- **Kick-off Meeting**
- Introduction to the Research - Practice - Venture Project approach
- Introduction to the work process of the project and meeting planning
- Introduction to the content and tasks of the project by Practice Partner
- Time planning and next steps.
- **Different meeting during the semester with academic coach and corporate partners. Appointments to be arranged.**
- **Final presentation meeting with academic coach and corporate partner.**

Literature

None specific

Additional information

The project meetings can be held in person or online. The planning of regular meetings is part of the project management skill to be trained during the project. Students take responsibility.

Examination information

Examination sub part/s

1. Examination sub part (1/2)

Examination modalities

Examination type	Written work
Responsible for organisation	decentral
Examination form	Written work
Examination mode	Digital
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Group work group grade
Weighting	80%
Duration	--

Languages

Question language: English
Answer language: English

Remark

Project Report (PowerPoint Slides)

Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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2. Examination sub part (2/2)

Examination modalities

Examination type	Presentation
Responsible for organisation	decentral
Examination form	Oral examination
Examination mode	Analog
Time of examination	Term time
Examination execution	Asynchronous
Examination location	On Campus
Grading type	Group work group grade
Weighting	20%
Duration	--

Languages

Question language: English
Answer language: English

Remark

Presentation (Executive Summary)

Examination-aid rule

Free aids provision

Basically, students are free to choose aids. Any restrictions are defined by the faculty members in charge of the examination under supplementary aids.

Supplementary aids

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Content

Project Report as PowerPoint (80%):

- Practical quality 20%
- Academic quality 20%
- Process management 20%
- Social competence 20%

Oral PowerPoint presentation of results to the client and the academic coach (20%)

Literature

no specific literature required

Please note

Please note that only this fact sheet and the examination schedule published at the time of bidding are binding and takes precedence over other information, such as information on StudyNet (Canvas), on lecturers' websites and information in lectures etc.
Any references and links to third-party content within the fact sheet are only of a supplementary, informative nature and lie outside the area of responsibility of the University of St.Gallen.

Documents and materials are only relevant for central examinations if they are available by the end of the lecture period (CW51) at the latest. In the case of centrally organised mid-term examinations, the documents and materials up to CW 43 (Freitag, 24. October 2025) are relevant for testing.

Binding nature of the fact sheets:

- Course information as well as examination date (organised centrally/decentrally) and form of examination: from bidding start in CW 34 (Thursday, 21 August 2025);
- Examination information (supplementary aids, examination contents, examination literature) for decentralised examinations: in CW 42 (Monday, 13. October 2025);
- Examination information (supplementary aids, examination contents, examination literature) for centrally organised mid-term examinations: in CW 43 (Freitag, 24. October 2025);
- Examination information (regulations on aids, examination contents, examination literature) for centrally organised examinations: two weeks before ending with de-registration period in CW 45 (Monday, 03. November 2025).